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DAPSTOM - An Integrated Database & Portal for Fish Stomach Records



Version 6.3

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Executive Summary

This report describes the latest efforts to update and expand the DAPSTOM database for fish stomach content records. In this latest iteration (Version 6.3) an additional 26,767 records for 122,137 individual predator stomachs have been added to the dataset (including 31,488 cod stomachs), bringing the total up to 283,121 records from 481,476 stomachs and 741 distinct research cruises/sampling campaigns. Data are available for 210 predator species and a huge swathe of the Northeast Atlantic, but particularly the North Sea, Celtic Sea, Irish Sea and area around Spitzbergen. Records span the period 1836 to 2023 and in this latest version particular emphasis has been placed on digitising historical datasets from Scottish waters, the Faroe Islands, the coasts of Norway, the Irish Sea and freshwater ecosystems. The DAPSTOM database encompasses individuals ranging in size from 0.1 cm (a herring larva) to 768 cm for a basking shark caught in 1947. New predator species added in this iteration include four freshwater fish species that are endemic to the British Isles, namely: the Gwyniad/Pollan (*Coregonus pennantii*), Ferox trout (*Salmo ferox*), Sonaghen (*Salmo nigripinnis*) and Gillaroo (*Salmo stomachicus*), as well as Vahl's eelpout (*Lycodes vahlii*) and chub (*Squalius cephalus*).

This update to DAPSTOM has been funded via the European Union under project “SC10 – Study on stomach content of fish to update databases and analyse possible changes in diet or food webs interactions” (EASME/EMFF/2018/011). This report describes efforts to substantially re-engineer the online portal and move away from Microsoft Access toward PostgreSQL. Postgres is a free and open-source relational database management system that can be linked to Cefas’ externally accessible server, as part of the ‘Cefas Open Science’ initiative. The intention is that the DAPSTOM Version 6.3 database will be made searchable and freely available to users via a dedicated R-shiny interface.

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1 Introduction

It has been five years since the last update to the DAPSTOM database (Version 5.5). This report describes efforts to add 26,767 additional records, but also to substantially re-engineer the online portal and move away from Microsoft Access toward PostgreSQL. Postgres is a free and open-source relational database management system that can be linked to Cefas' externally accessible server, as part of the 'Cefas Open Science' initiative. The intention is that the DAPSTOM Version 6.3 database will be made searchable and freely available to users via a dedicated R-shiny interface.

This update to DAPSTOM 6.3 has been funded via the European Union under "Lot 1: Scientific advice in support of the CFP in the Baltic Sea and North Sea EU waters", and specifically project "SC10 – Study on stomach content of fish to update databases and analyse possible changes in diet or food webs interactions" (EASME/EMFF/2018/011). Under this programme, new samples of cod and whiting have been collected from the Baltic Sea along with mackerel, whiting, and grey gurnard from the North Sea. Additionally, further improvements have been funded within the DAPSTOM database. The contract requires that the DAPSTOM database will contribute an additional 18,000 stomach content records (as a minimum) until 2021. Newly developed linear mixed effect models are being used in DAPSTOM to provide estimates of prey biomass in predator samples.

1.1 Database structure and design

As part of the EU 'Lot' project MARE/2012/02 in 2014, the DAPSTOM database was redesigned and substantially altered. Further reconstruction and refinement has been carried out under SC10 (EASME/EMFF/2018/011). The basic relational structure has been retained for version 6.3 (figure 1), although 'predators' and 'prey' in their stomachs have now been split into two separate tables, noting that there can be multiple prey records in the 'prey' table for each individual (or pooled) predator in the 'predator' table. In addition, a new 'prey qualifier' look-up table has been added, linked to the 'prey' table to differentiate between prey life stages, e.g. eggs, larvae, adults (see table 1). Prey names, as written in the original source material are retained in the 'prey table', but a TSN identifier is assigned to correct these names to an internationally recognised taxonomic scheme. TSN numbers are used as the linking variable to the 'prey taxonomy' table. Taxonomic Serial Numbers (TSN codes) are issued by the Integrated Taxonomic Information System (ITIS), a partnership formed in 1996 as an interagency group within the US federal government. TSN numbers were chosen to provide the primary taxonomic architecture for DAPSTOM in preference to other schemes (such as the WoRMS - World Register of Marine Species) because ITIS-TSN includes terrestrial, freshwater and marine organisms, whereas other schemes have gaps. This being said, it was necessary (in a few cases) to devise our own DAPSTOM/TSN codes for empty stomachs and inanimate objects etc. (see table 2).

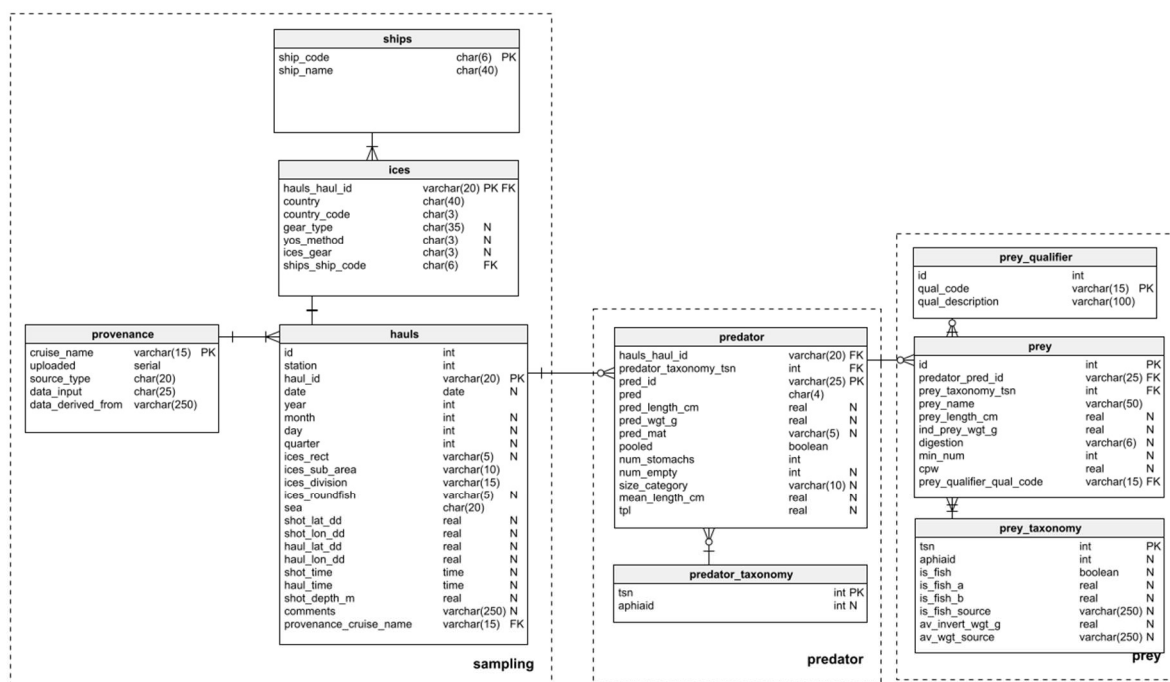


Figure 1. Structure of the revised **DAPSTOM 6.3** relational database, including a list of the fields included (for a full description see Appendix 1).

Table 1. Prey qualifier codes, used to differentiate different life stages.

Qualifier code	Qualifier description
Q1	None
Q2	Larvae
Q3	Elvers
Q4	Eggs
Q5	Adult
Q6	Adult-female
Q7	Adult-male
Q8	Copepodite
Q9	Copepodite 1-3
Q10	Copepodite 4-5
Q11	Nauplii
Q12	Pupae
Q13	Siphons (mollusc)
Q14	Ephyra
Q15	Nymphs

Q16	Imago
Q17	Megalopae
Q18	Juvenile
Q19	Cypris (barnacle)
Q20	Zoea
Q21	Parasite

Table 2. Assigned TSN codes for prey items in the DAPSTOM database that were not available via the Integrated Taxonomic Information System (ITIS).

TSN code	Description
-99901	Empty
-99902	Regurgitated/everted
-99903	Unreadable
-99904	Unidentifiable/digested remains
-99905	Wood
-99906	Stones/rock
-99907	Sand/grit
-99908	Mud
-99909	Metal
-99910	Paper rubbish
-99911	Plastic rubbish
-99912	Fish-hooks
-99913	Unidentified benthic invertebrates
-99914	Other (unidentified) invertebrates
-99915	Unidentified invertebrate eggs

1.2 New datasets added in version 6.3

Data from 127 additional sampling campaigns were added to the database as part of this most recent update. Full details can be obtained by consulting the 'PROVENANCE' table within the database. Some datasets were derived from scientists log-books or reports contained within the Cefas archive, others were donated by 'partners' or were digitised from published peer-reviewed papers, with specific relevance to the British Isles.

Donated datasets include those provided by Adriaan Rijnsdorp (Wageningen Marine Research) as part of EU BENTHIS project (see Rijnsdorp & Vingerhoed 2001) and including cruises TRIDENS-01-1996, ISIS-01-1996, ISIS-02-1996, ISIS-03-1996 and U383-01-1996. An electronic dataset from the French EVHOE survey in the Celtic Sea (cruises THALASSA-2014 and THALASSA-2015) was made available by colleagues from the IFREMER laboratory in Lorient (see supplementary material in Marianne et al. 2022).

Recent datasets (2017-2023) include those collected as part of the annual Clean Seas Environmental Monitoring Programme (CSEMP), provided by Manuel Nicolaus and Freya Goodsir (Cefas). These comprise data on dab stomach contents, collected at specific trawl localities around the North Sea coast (2017, 2019, 2021) or English Channel, Bristol Channel and Irish Sea (2018, 2020, 2023). In addition, stomach datasets for the Northeast coast of England (dab and plaice) were provided by Georgina Hunt and resulted from a joint PhD between University of Newcastle and Cefas (see cruise "HUNT-2019") (Hunt 2021; Hunt et al. 2023). Data concerning pelagic fish species (anchovy, horse mackerel, sprat and sardine) were provided by Roweena Patel at University of Reading (cruise "END15-19"), based on samples from Cefas' PELTIC survey in October 2019 (Patel et al. 2023).

Historical datasets identified in the Cefas archive, and subsequently digitised include:

- Data digitised from spring-bound internal report found at Pinbush: Bellwood, D.R. (1978) The ecology of 1-group sole with particular reference to food and feeding. Industrial Training Report. Ministry of Agriculture, Fisheries & Food (MAFF), Directorate of Fisheries. 17pp.
- Data digitised from: Garrod, C. and Harding, D. (1981) Predation by fish on the pelagic eggs and larvae of fishes spawning in the west central North Sea in 1976. ICES CM 1981/L:11.
- Data digitised from: Last, J.M. (1979) The food of larval turbot *Scophthalmus maximus* L. from the west central North Sea. J. Cons. int. Explor. Mer, 38(3): 308-313.
- Data digitised from a folder full of raw data located during rationalisation of Pinbush. Also see: Macer, C.T. (1977) Some aspects of the biology of the horse mackerel [*Trachurus trachurus* (L.)] in waters around Britain. Journal of Fish Biology, 10: 51-62.
- Data digitised from: Pentelow, F.T.K., Southgate, B.A., Bassindale, R. (1933) II The proportion of the sexes and the food of smoults of salmon and sea trout in the Tees Estuary. Fishery Investigations, Series I., Vol III, Number 4, 11-14.
- Summary data digitised from a folder of Irish Sea records provided by Mike Armstrong (raw data lost; partially digitised in DAPSTOM 1.5 and 3.6).

Datasets digitised directly from published papers and reports included:

- Information on commercial fish species in the Norwegian Trench (Albert 1993; Albert 1994a,b; Albert 1995)
- Information on fish species from the Norwegian fjords and Skagerrak coast, both adults and larval fish (Bjorke 1976; Hop et al. 1992; Mattson 1981; Mattson 1990; Mattson 1992; Nedreaas 1985; dos Santos & Falk-Petersen 1989; Fjøsne & Gjøsæter 1996; Sunnana 1984).

- Information on commercial fish species around the Faroe Islands (Magnussen 2011; Jacobsen & Hansen 1996; Rae 1967b; Hislop & Youngson 1984).
- Information on fish species around Scotland – mostly taken from reports of the Department of Agriculture and Fisheries for Scotland (Ogilvie 1927; Ogilvie 1938; Rae 1956; Rae 1967; Rae & Shelton 1982; Daly et al. 2001; Mendonca et al. 2007; Sarno et al. 1994; Sayer et al. 1995; Robb & Hislop 1980; Robb 1981; Economou 1991; Klimpel et al. 2003).
- Information on freshwater fish species in Scotland, Wales, Ireland and England (Adams & Maitland 1998; Ferguson & Mason 1981; Pentelow 1932; Arawomo 1981; Garner 1996; Garner 1996; Hutchinson 1983; Mann et al. 1997; Pentelow 1933; Maitland 1965; Sinha, & Jones 1967; McGoran et al. 2016; Carpenter 1940; Jones 1951; Dunn 1954; Ball 1961; Graham & Jones 1962; Weatherly 1987; Hellawell (1971, 1972, 1974)).
- Flatfish and inshore species in the North Sea, especially off the Netherlands and French coast (Thijssen et al. 1974; Wetsteijn 1981; Beyst 1999; Daan 1973; De Grooy 1971; Lagardere 1987; Maes & Ollevier 2002).
- Elasmobranchs in the Irish Sea and North Sea (Ajayi 1982; Holden 1966; Holden & Tucker 1974; Lyle 1983).
- Miscellaneous inshore coastal fish in Wales, SW England, Scotland and Ireland (Seisay 2001; Rogers, 1991; Berg 1979; Badsha & Sainsbury 1978; Daly et al. 2001; Deady & Fives 1995a,b; Tully & O Ceidigh 1989; Liao 1999; Menon 1950).
- Stomach contents of immature bluefin tuna in the Bay of Biscay (Ortiz de Zarate & Cort 1986).

1.3 More complete data on average prey weights (invertebrates)

In preparing a previous version of the DAPSTOM database (Version 4.9), under the auspices of the EU MARE/2012/02 project, a major task involved the construction of a new 'look-up table' of average prey wet weight, that was further refined and expanded in version 5.5. A particularly important source of 'fresh weight' estimates was the Cefas UNICORN database of benthic surveys. Average wet weight of individuals by species were calculated, and this provided coverage for most benthic invertebrates encountered in fish stomachs. With regard to planktonic invertebrates, information on various macroplankton types (jellyfish, chaetognaths, euphausiids, mysids, ctenophores, larvaceans etc.) was derived from Moriarty et al. (2013), whereas information on mesozooplankton types (copepods, cladocerans etc.) was provided by Dr Sophie Pitois of Cefas and was based on the volume of different species, converted to wet biomass (see Pitois and Fox 2006). Estimates of 'fresh weight' for insects were primarily obtained from Studier & Sevic (1992) and from Chown et al. (2007).

In collating Version 6.3 of the DAPSTOM database, many new prey items were mentioned/detected in the original sources. Again, extensive use was made of 'fresh weight' estimates for benthic invertebrates from the Cefas UNICORN database.

In some cases, no relevant information was available in the sources cited above, or the particular prey type comprised a higher-level taxonomic category (e.g. polychaetes). In this case, we made use of observed weights of individuals actually observed in stomachs from the DAPSTOM database itself, where these existed. This was especially true for plant material (seaweeds and terrestrial plants) as well as colonial animals such as bryozoans, sponges and soft corals etc. For phytoplankton and micro-organisms (such as foraminifera) a nominal small value of 0.00001 g per individual was assumed.

1.4 More complete data on average prey weights (fish and cephalopods)

For all fish species observed as prey, average prey weight estimates were not used. Rather weight-length exponents (a , b) were included (in the 'prey taxonomy' table), mostly derived from the Cefas report by Silva et al. (2013), and these were in-turn used to derive prey weight, based on the length of the predator. Within the 'predator' table, a field called 'TPL' (Theoretical Prey Length) had previously been included based on the observed relationship between predator length and prey length. Individual species' relationships ($n=22$) from Pinnegar et al. (2003) and Scharf et al. (2000) were averaged, to create an overall relationship:

$$\text{prey length} = (0.2057 \times \text{predator length}) + 1.618$$

Weight-length exponents (in the 'prey taxonomy' table) were used with estimates of TPL (in the 'predator' table) to yield the wet weight of fish prey items.

In addition to fish, weight-length relationships were also available for cephalopods (squid, octopus and cuttlefish), most notably from Merella et al. (1997) and Emam et al. (2014). For the

frog *Rana temporaria* exponents were obtained from Gibbons and McCarthy (1984), for harbour porpoise *Phocoena phocoena* from Kastelein and van Battum (1990) and for birds (*Anthus pratensis* and *Uria aalge*) from Victoria et al. (2013). As part of DAPSTOM 6.3, notable new prey items included harbour seal (eaten by a 75.7cm spurdog in 1966) and water vole (eaten by two chub in 1964). Weight-length exponents for these species were taken from Markussen et al (1989) and Pisanu et al. (2003) respectively.

2 Species, spatial and temporal coverage

2.1 Species Coverage

The most recent version of the DAPSTOM dataset (Version 6.3, collated in August 2023) includes 283,121 records derived from 741 distinct research cruises, spanning the period 1836-2023. The database contains information from 481,476 individual predator stomachs and 210 predator species. As such, this represents one of the largest and most diverse compilations of marine food-web data anywhere in the world.

In this most recent upload, an additional 26,767 records were added to the database from 122,137 individual stomachs. The fact that the number of stomachs is substantially higher than the number of records reflects the fact that large quantities of ‘pooled’ data were digitised. For example, 4210 records were obtained where cod was identified as the predator, but these were derived from 31,488 stomachs (figure 2). Similarly, 570 records were digitised for salmon, from 7543 individual stomachs. By contrast, the situation was reversed for dab, plaice and horse mackerel, where the number of stomachs exceeded the number of records (dab: 4291 records, 4071 stomachs; plaice: 3948 records, 3158 stomachs; horse mackerel: 530 records, 280 stomachs) and this reflects the fact that for these species large quantities of recent data were uploaded, collected at the individual stomach level (e.g. from the Cefas CSEMP cruises, from Hunt et al. 2023, Rijnsdorp & Vingerhoed 2001; Patel et al. 2023 and Macer 1977).

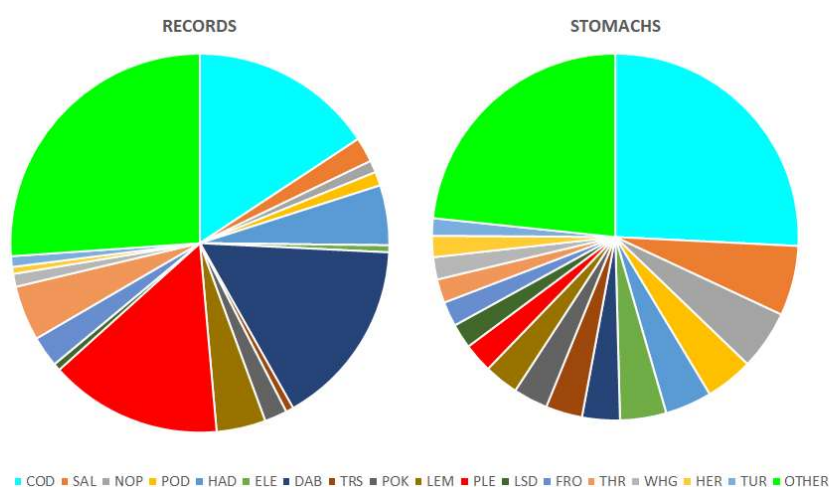


Figure 2: Composition of the newly added datasets by predator species, according to (a) the number of records, and (b) the number of stomachs examined [for 3 letter predator codes see Appendix 2].

Figure 3 shows that for the DAPSTOM Version 6.3 database as a whole, 26.8% of all records (75,923) relate to cod and 16.9% (47,929) relate to whiting. However, the importance of these species in terms of number of stomachs is much reduced (16.4% and 7.5% respectively), whereas plaice and herring represent a greater proportion of the stomachs examined (9.3% and 16.1% respectively) than of the number of records (6.8% and 2.8% respectively). For 18 species (out of 212) in the database, only one individual stomach was examined.

New predator species added in this iteration include four freshwater fish species that are endemic to the British Isles, namely: the Gwyniad (*Coregonus pennantii*), Ferox trout (*Salmo ferox*), Sonaghen (*Salmo nigripinnis*) and Gillaroo (*Salmo stomachicus*), as well as Vahl's eelpout (*Lycodes vahlii*) and chub (*Squalius cephalus*). The primary source materials for these new additions were:

- Gwyniad [WALES-1951, WALES-1952, WALES-1961, WALES-1962] from Dunn (1954) and Haram & Jones (1971), Llyn Tegid (Bala Lake), Merionethshire, Wales.
- Ferox trout, Sonaghen and Gillaroo [FERGUSON-1981] from Ferguson & Mason (1981), Lough Melvin, Ireland.
- Vahl's eelpout [ALBERT-1987A] from Alber (1993) in the Norwegian Deep.
- Chub [GARNER-1993, GARNER-1994, HELLAWELL-1964], from Garner (1996) and Hellawell (1971, 1972, 1974), River Great Ouse, England.

For a full breakdown of the number of records and individual stomachs per species, see Appendix 2.

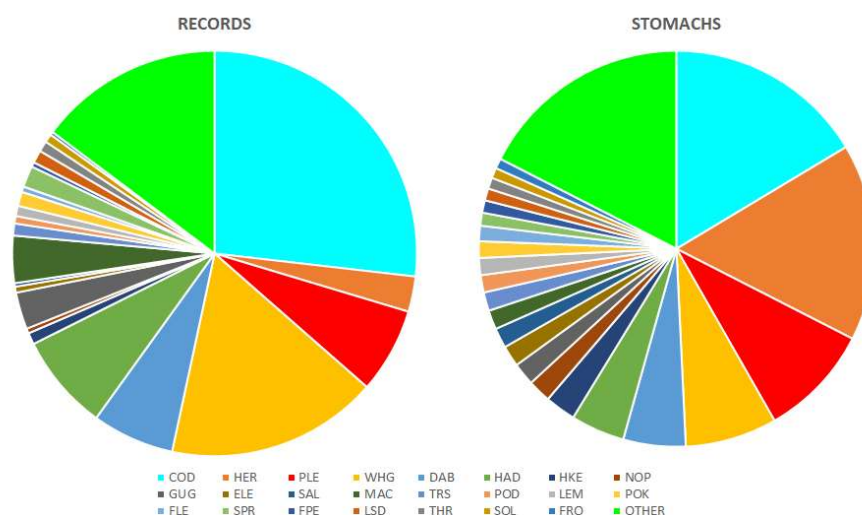


Figure 3: Composition of the DAPSTOM 6.3 database by predator species, according to (a) the number of records, and (b) the number of stomachs examined [for 3 letter predator codes see Appendix 2].

2.2 Spatial Coverage

In this latest iteration of the DAPSTOM database (version 6.3), the largest proportion of newly added data (49%, 13,217 records) were derived from the North Sea and this is also the case when expressed in terms of number of stomachs (53%, 64,680 stomachs). 13% of all records (3,576) were derived from the Irish Sea, but only 6% of all stomachs (7,365). Similarly, large quantities of data (11%, 1,257 records) were digitised from the Celtic Sea (including the Bristol Channel), although these records were derived from a smaller number of stomachs (5%, 5,831 stomachs). By contrast, a relatively small proportion of records were obtained for freshwater predatory fish (8%, 2,220 records), but these were derived from a relatively large number of stomachs (12%, 14,675 stomachs).

As is apparent from Table 3, the updated DAPSTOM database (Version 6.3) includes information from sites all over the Northeast Atlantic. However, half (52.8%) of the records (52.6% of stomachs) relate to the North Sea, given that this has continued to be the main focus of survey work at Cefas/MAFF in Lowestoft for the past 120 years. Relatively large numbers of records have also been digitised for the Irish Sea (10.7% records, 9.5% stomachs), Celtic Sea (8.8% records, 4.4% stomachs), as well as the area around Spitzbergen (9.0% records, 4.0% stomachs), where a MAFF survey vessel operated from 1949 to 1977 (see Townhill et al. 2015). Freshwater fish species represent only a small proportion of all records (2.6%) and stomachs (6.5%) in the database.

Table 3: Number of records and stomachs examined, by geographic area.

Sea	ICES Sub-Areas	RECORDS	% RECORDS	STOMACHS	% STOMACHS
Baltic	IIId (22, 24, 25, 26, 28)	628	0.2	3683	0.8
Barents Sea	Ib	3127	1.1	2662	0.6
Biscay	VIIIa-d	2067	0.7	1690	0.4
Bristol Channel	VIIIf	2887	1.0	6959	1.4
Celtic Sea	VIIg-j	24796	8.8	21095	4.4
Channel	VIIId,e	11654	4.1	30982	6.4
E Greenland	XIVa,b	3154	1.1	1288	0.3
Faroes	Vb	3106	1.1	7452	1.5
Freshwater	NA	7370	2.6	31567	6.5
Iceland	Va	2253	0.8	1474	0.3
Irish Sea	VIIa	30353	10.7	45703	9.5
Skagerak & Kattegat	IIIa + IIIb (21, 23)	2436	0.9	7638	1.6
North Atlantic	XII	18	0.0	9	0.0

North Sea	IVa,b,c	149545	52.8	253679	52.6
Norwegian Sea	IIa	7155	2.5	15376	3.2
Spitzbergen	IIb	25582	9.0	19229	4.0
W Ireland	VIIb,c	2791	1.0	15541	3.2
W Scotland	VIa	3782	1.3	16373	3.4
Labrador	NAFO 2 (2H, 2J)	417	0.1	298	0.1

2.3 Temporal Coverage

The earliest records in the DAPSTOM database are for 1836 and were digitised from Thompson (1838) in *Annals of Natural History*. The most recent record is from the RV *Cefas Endeavour* 'END01-23' cruise in January 2023. Figure 4 shows that data have been digitised from every decade after 1830. From the 1920s (6,737 records, 54,497 stomachs) onwards the number of fish stomachs examined increased rapidly. Fewer stomachs were sampled in the 1940s, because activity ceased during the second World War, however sampling resumed quickly in the 1950s (figure 4 31,636 records, 55,641 stomachs). Many thousands of stomachs have been digitised for the 1970s (23,676 records; 73,124 stomachs) and progressively fewer stomachs have been sampled in decades since. It should be noted however, that additional stomach sampling effort was expended in 1981 and 1991 as part of the ICES 'Year of the Stomach' campaigns, but these data are not included in the DAPSTOM database in order to avoid duplication with the open-access sources available through ICES (see www.ices.dk/marine-data/data-portals/Pages/Fish-stomach.aspx). The most recent iteration of the DAPSTOM database added large quantities of data from the 1960s and 1970s in particular (24,780 and 30,995 stomachs respectively). In the current decade (2020-2023) 773 records (541 stomachs) were derived from the annual CSEMP (Clean Seas Environmental Monitoring Programme) monitoring cruises [END18-20, END06-21, END01-23].

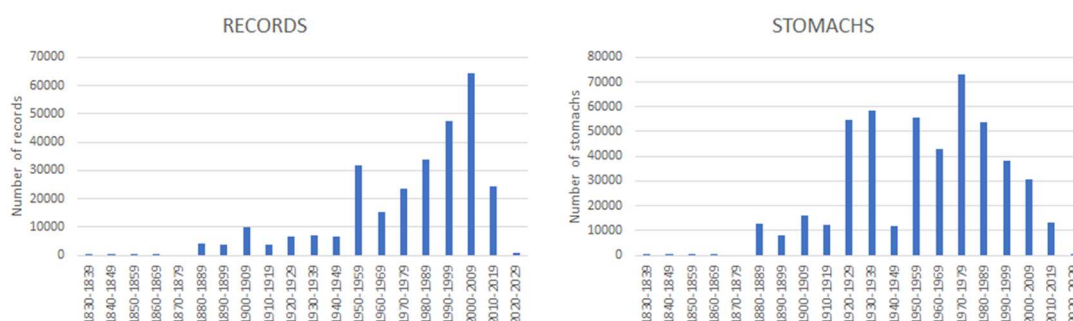


Figure 4. Temporal coverage of database records and stomachs sampled, within the DAPSTOM 6.3 database.

3 Discussion – how the database can be used

3.1 Caveats and capabilities

As a result of efforts to estimate ‘calculated prey weight’ (CPW) it is now possible to compare the nominal diet composition based on both number of prey items consumed, but also the weight (mass) of prey consumed. For example, figure 4 shows the results for dab and plaice >10cm total length in the North Sea. From this figure it is clear that diet composition of dab includes large numbers of bivalve molluscs (23%) and ophiuroids (16%), whereas these prey types are less important in terms of weight (13 and 12% respectively). By contrast Anomurans (hermit crabs and squat lobsters), teleosts and brachyuran crabs represented only a small proportion of prey in terms of numbers consumed (7, 7 and 6%) but they are very important in terms of biomass (22, 13 and 14% respectively). In plaice – bivalve molluscs are important prey items both in terms of number and weight of prey items consumed (72 and 70% respectively), but polychaetes were more important in terms of numbers in comparison with biomass (16% compared to 7%) whereas fish (teleosts) were less important in terms of numbers in comparison with biomass (2% compared to 10%). These results are largely indicative and are very sensitive to the assumed average size of an individual prey organism – however the estimates should prove useful for future multispecies modelling projects (e.g. Bentley et al. 2019).

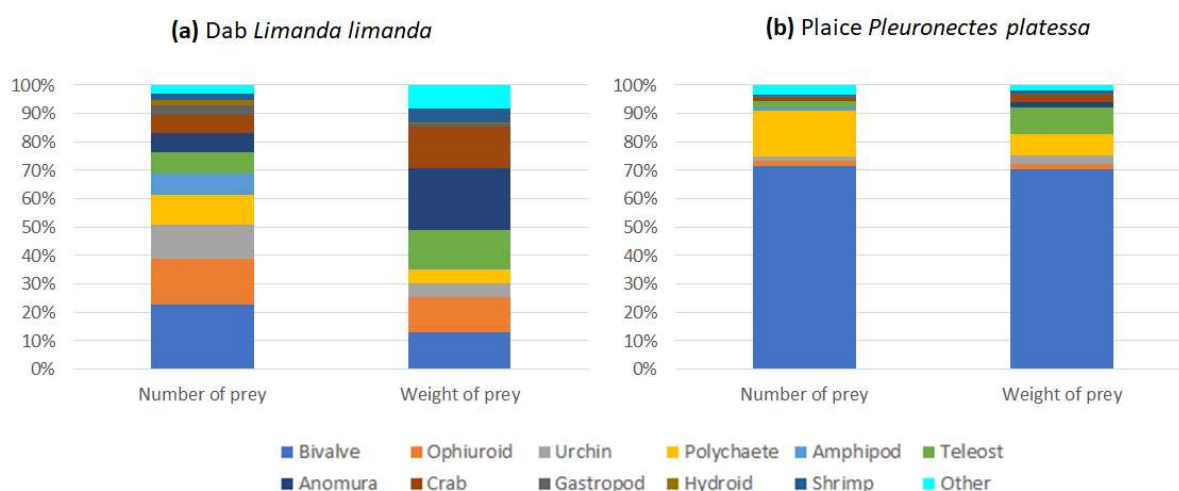


Figure 5. Diet composition of (a) dab *Limanda limanda* and (b) plaice *Pleuronectes platessa* based on the number and estimated biomass (wet weight) of prey items consumed.

A major limitation of the DAPSTOM Version 6.3 dataset is that it comprises a mixture of ‘pooled’ information together with data collected from individual fish. Sometimes only information on the number of stomachs containing a particular prey item was available (i.e. ‘frequency of occurrence’), rather than the actual number of a particular prey item. Hence in any data extraction, outputs should be viewed as providing information on the ‘minimum number’ of prey items consumed. This would have little impact in predator species that consume large prey items (e.g. fish feeders), and in most of the newer datasets submitted, but it could mean that in certain older datasets, the total number of prey items (and by extension the aggregate weight of prey items consumed) in plankton-eating species such as herring could be underestimated.

An example of such data added in this most recent iteration includes that from Ogilvie (1927) on herring and diets of larval haddock from Ogilvie (1938) around Scotland.

Unlike the ICES 'Year of the Stomach' dataset, the DAPSTOM database includes considerable information on the prey consumed by fish larvae as well as adults. In effect, this means that any data extraction that does not take account of predator size (and for example exclude individuals smaller than 10cm) could yield a diet composition with unexpectedly high prominence of planktonic prey items such as copepods. The DAPSTOM database encompasses individuals ranging in size from 0.1cm (a herring larva) to 768 cm for a basking shark caught in 1947.

3.2 How has DAPSTOM been used since 2019?

Since 2019 the DAPSTOM database and its associated reports have been cited in many peer-reviewed publications. It is increasingly being viewed as a major resource for ecosystem modelling and for macro-ecological studies. In particular, the database has been used to parameterise multispecies fisheries models and to develop ecosystem indicators, with a focus on 'food-webs', as well as basic investigations into the underlying biology of particular species.

The original DAPSTOM database (version 1) was created in order to help parameterise multispecies and ecosystem models for the seas surrounding the British Isles. This activity remains central to database developments and a number of modelling papers have been published since 2019 that make use of this resource. Spence et al. (2021) implemented a multispecies assessment for the Irish Sea, fitting a model of eight key stocks directly to catch and survey data. Using the DAPSTOM database the authors (Spence et al. 2021) developed a 'suitability matrix'. If any of the sample stomachs of species *X* contained species *Y* then species *X* was deemed to be capable of predating on species *Y* (suitability = 1), otherwise it was assumed not to eat species *Y* (suitability = 0). Estimates of natural mortality showed modest year to year variability, suggesting that single species assessments may be appropriate for short term tactical management. This method shows great promise for making multispecies assessments as a complement to existing assessments because of the rapid turnaround time and ability to respond at meetings to the requests of stakeholders (Spence et al. 2021).

Bentley et al. (2019) used Linear Inverse Modelling with an Ecopath model of the Irish Sea to investigate how the incorporation of uncertainty in dietary data can strengthen inferences based on model-derived Ecological Network Analysis (ENA) indicators. Diet data for fish functional groups were obtained from DAPSTOM. The default data format in DAPSTOM is counts of prey per predator, which were converted into weight (kg) using average estimates for each prey species. Average diets, along with plausible ranges were generated for each fish functional group based on annual data (1960–2017). Traditionally, Ecopath diet matrices are altered *ad-hoc* during model-balancing. In this study the authors (Bentley et al. 2019) used the stomach records to establish a data-based parameter field to move within when balancing the model (Bentley et al., 2018).

Food web indicators can reveal how ecosystems are responding to environmental change and anthropogenic pressure in a way that cannot be inferred from studying habitat, species or

assemblages alone. The DAPSTOM database in its entirety has been used to define feeding guilds, the status of which can be tracked over time and space, as part of monitoring commitments under the EU Marine Strategy Framework Directive (see Thompson et al., 2020) and the Convention for the Protection of the Marine Environment of the North-East Atlantic (the “OSPAR Convention”). Thompson et al. (2020) found evidence for seven distinct feeding guilds in the North Sea, based on a dataset containing 8,092 unique predator–prey interactions from 415,294 fish stomachs. Guild biomasses remained relatively constant through time (1985–2014) at the North Sea-level (Thompson et al. 2020), but the species composition within these guilds changed appreciably over time.

Townhill et al. (2021) assembled a new dataset of stomach content records for Atlantic cod (*Gadus morhua*) in the Barents Sea by combining historic (DAPSTOM) data collected by UK scientists in the 1930–1964 (see figure 6), with more recent data collected by Norwegian and Russian scientists. As a top predator, cod plays a key role in the Barents Sea ecosystem. The species has a broad diet consisting mainly of crustaceans and teleost fish. Visual analysis revealed that cod diets have changed considerably from the start of the dataset in the 1930s to the present day. There was a large proportion of herring in the diets in the 1930s, whereas in more recent decades capelin, invertebrates and other fish dominate. There are also significant interannual asynchronous fluctuations in prey, particularly capelin and euphausiids (Townhill et al. 2021).

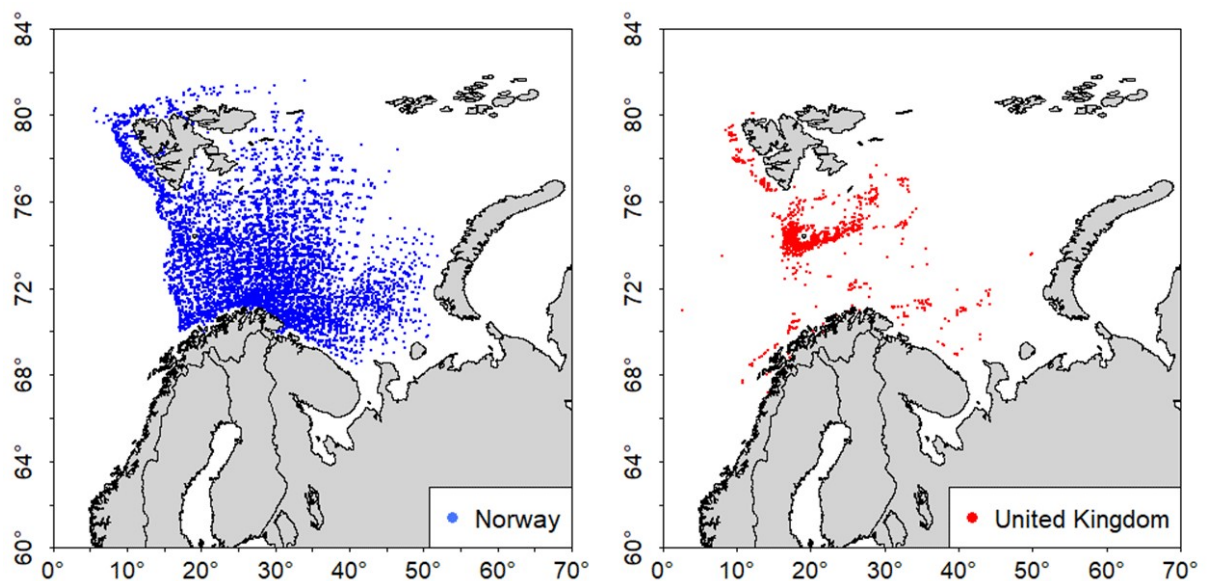


Figure 6. The location of cod stomach samples taken in the Barents Sea by Norway (1984–2018) and the United Kingdom (1930–1964) [from Townhill et al. 2021].

Hunt (2021) set out to investigate multi-decadal trends in the diet composition of six demersal and benthivorous fish predators in the western North Sea (cod, whiting, haddock, grey gurnard, plaice, dab). This PhD chapter [chapter 4] utilised DAPSTOM (version 5.5) data together with the ICES ‘Year of the Stomach’ dataset, to identify shifts in the importance of particular benthic prey items throughout 20th century. Over the period 1896–2015, there were substantial changes in the diet composition of prey consumed by all six predators >20 cm between April–September. For plaice, bivalves (mostly *Macridae* sp., *Ensis* sp., *Spisula subtruncata* and *Tellina*

tenuis) were major prey in the early to mid-20th century, dominating the diet in the 1950s in terms of number and mass. However, bivalves declined in subsequent decades and polychaetes became more prominent in plaice diet from the 1960s onwards. Bivalves also declined in importance by number in dab diet from 35.8% to 12.4% between the 1900s and 2010s. The proportion of echinoderms in flatfish diets increased in importance from the early 1900s to 2000s. Results are interpreted in context of human pressures, predominantly fisheries, climate change and eutrophication. This work underlines the value of historical diet data to examine long-term changes in marine ecosystems. Hunt et al. (2023) (Chapter 5 within Hunt 2021) also collected contemporaneous stomach content data for plaice and dab across multiple locations on the Northumberland coast. These data, together with outputs from stable isotope analysis of fish tissues (muscle and liver) were used to characterise temporal and spatial variability in diets and have since been uploaded as part of DAPSTOM version 6.3.

Small pelagic fish (SPF) play an important ecological role by facilitating energy transfer in marine ecosystems, from lower to higher trophic levels. Patel et al. (2023) investigated the seasonal diet composition of 6 SPFs in the Celtic Seas ecoregion (anchovy, herring, horse mackerel, mackerel, sardine, and sprat), identified diet overlap, and determined species selectivity towards particular prey sizes. Combining historical (DAPSTOM version 5.5) and new observations (now included in DAPSTOM version 6.3) on fish-stomach contents, diet-overlap indices were computed to quantify potential seasonal intra- and inter-specific overlap. Diet overlaps among species changed between seasons; in particular, sprat and anchovy between spring and autumn, and mackerel between spring, summer, and winter. SPF species primarily consumed calanoid copepods, with herring and horse mackerel having more selective diets compared to mackerel, anchovy, and sardine (Patel et al. 2023).

Barrett et al. (2022) analysed historical fish stomach records from DAPSTOM and revisited literature at local and regional level to determine the importance of cephalopods to the diets of 26 ecologically important finfish in the Northeast Atlantic. The authors concluded that, in contrast to most other large marine ecosystems, cephalopods found in the Greater North Sea and the Celtic Seas regions appear to contribute only a small fraction to the diets (found in the stomachs of ~14% of specimens among some species, but generally only 1–3% in most species), though their role as predator may be important and require further investigation (Barrett et al. 2022).

3.3 Developing a new user interface with R-shiny

[To be completed by Murray]

3.4 Looking to the future

Data collection for Version 7 of the DAPSTOM database is already underway. New data continues to be collected as part of the annual CSEMP programme. In addition, a substantial quantity of un-digitized historical data from 1951-1953 has been sent to Cefas by University of

Liverpool, that derives from Scottish surveys in the Moray Firth, and was previously analysed by Frid & Hall (1999) and Frid et al. (2001). Partners at JNCC have agreed to work with Cefas to collect new stomach data for offshore Highly-Protected Marine Areas (HPMAs), in particular North East of Farnes Deep and off Dolphin Head.

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[HELP MINTING A DOI]

5 References

- Adams, C.E. and Maitland, P.S. (1998) The Ruffe population of Loch Lomond, Scotland: Its introduction, population expansion and interaction with native Species. *J. Great Lakes Research*, 24(2): 249-262.
- Ajayi, T.O. (1982) Food and feeding habits of Raja species (Batoidei) in Camarthen Bay, Bristol Channel. *J. Mar. Biol. Ass. U.K.* 62: 215-223.
- Albert, O. T. 1993. Distribution, population structure and diet of silvery pout (*Gadiculus argenteus thori* J. Schmidt), poorcod (*Trispterus minutus minutus* (L.)), four-bearded rockling (*Rhinonemus cimbricus* (L.)), and Vahl's eelpout (*Lycodes vahlii gracilis* Reinhardt) in the Norwegian Deep. *Sarsia*, 78: 141–154.
- Albert, O.T. (1994) Biology and ecology of Norway pout (*Trisopterus esmarki* Nilsson, 1855) in the Norwegian Deep. *ICES Journal of Marine Science*, 51: 45-61.
- Albert, O.T. (1994) Ecology of haddock (*Melanogrammus aeglefinus* L.) in the Norwegian Deep. *ICES Journal of Marine Science*, 51: 31-44.
- Albert, O.T. (1995) Diel changes in food and feeding of small gadoids on a coastal bank. *ICES Journal of Marine Science*, 52: 873-885.
- Arawomo, G.A.O. (1981) The food of juvenile trout, *Salmo trutta* L., in Loch Leven, Kinross, Scotland. *Hydrobiologia*, 79: 105–112.
- Badsha, K.S. & Sainsbury, M. (1978) Aspects of the biology and heavy metal accumulation of *Ciliata mustela*. *Journal of Fish Biology*, 12: 213-220.
- Ball, J.N. (1961) On the food of the brown trout of Llyn Tegid. *Proceedings of the Zoological Society of London*, 137(4): 599-622.
- Barrett, C.J., Barry, P., MacLeod, E., Stott, S., Vieira, R., Laptikhovsky, V. (2022) The importance of cephalopods in the diet of fish on the northwest European shelf. *ICES Journal of Marine Science*, 79: 1675–1686.
- Bellwood, D.R. (1978) The ecology of 1-group sole with particular reference to food and feeding. Industrial Training Report. Ministry of Agriculture, Fisheries & Food (MAFF), Directorate of Fisheries. 17pp.

- Bentley, J. W., Hines, D., Borrett, S., Serpetti, N., Fox, C., Reid, D. G., and Heymans, J. J. 2019. Diet uncertainty analysis strengthens model-derived indicators of food web structure and function. *Ecological Indicators*, 98: 239–250.
- Bentley, J. W., Serpetti, N., Fox, C. J., Reid, D., and Heymans, J. J. 2018. Modelling the food web in the Irish Sea in the context of a depleted commercial fish community. Part 1: Ecopath Technical Report. Scottish Association for Marine Science Report, 294. 147 pp.
- Berg, J. (1979) Discussion of Methods of Investigating the Food of Fishes, with Reference to a Preliminary Study of the Prey of *Gobiusculus flavescens* (Gobiidae). *Marine Biology*, 50: 263–273.
- Beyst, B., Cattrijsse, A., Mees, J., B., (1999) Feeding ecology of juvenile flatfishes of the surf zone of a sandy beach. *Journal of Fish Biology*, 55: 1171–1186.
- Bjorke, H. (1976) Food and feeding of young herring larvae of Norwegian spawners. ICES CM 1976/H:36.
- Carpenter, K.E. (1940) The feeding of salmon parr in the Cheshire Dee. *Proceedings of the Zoological Society of London*, A110 (1-2): 81–96.
- Chown, S.L., Marais, E., Terblanche, J.S., Klok, C.J., Lighton, J.R.B. and Blackburn, T.M. (2007) Scaling of insect metabolic rate is inconsistent with the nutrient supply network model. *Functional Ecology*, 21, 282–290.
- Daan, N. (1973) A quantitative analysis of the food intake of North Sea cod, *Gadus morhua*. *Netherlands Journal of Sea Research*, 6 (4): 479–517.
- Daly, H.I., Pierce, G.J., Santos, M.B., Royer, J., Cho, S.K., Stowasser, G., Robin, J.-P., Henderson, S.M. (2001) Cephalopod consumption by trawl caught fish in Scottish and English Channel waters. *Fisheries Research*, 52:51–64.
- Deady, S. and Fives, J.M. (1995) Diet of Ballan wrasse, *Labrus bergylta*, and some comparisons with the diet of corkwing wrasse, *Crenilabrus melops*. *Journal of the Marine Biological Association of the United Kingdom*, 75: 651–665.
- Deady, S. and Fives, J.M. (1995) The diet of Corkwing wrasse, *Crenilabrus melops*, in Galway Bay, Ireland, and in Dinard France. *Journal of the Marine Biological Association of the United Kingdom*, 75: 635–649.
- De Grooy, S.J. (1971) On the interrelationships between morphology of the alimentary tract, food and feeding behaviour in flatfishes (Pisces: Pleuronectiformes). *Netherlands Journal of Sea Research*, 5(2): 121–196
- dos Santos, J. & Falk-Petersen, S. (1989) Feeding ecology of cod (*Gadus morhua* L.) in Balsfjord and Ullsfjord, northern Norway, 1982–1983. *J. Cons. int. Explor. Mer*, 45: 190–199.
- Dunn, D.R. (1954) The feeding habits of some of the fishes and some members of the bottomfauna of Llyn Tegid (Bala Lake), Merionethshire. *Journal of Animal Ecology*, 23 (2): 224–233.
- Economou, A.N. (1991) Food and feeding ecology of five gadoid larvae in the northern North Sea. *J. Cons. int. Explor. Mer*. 47: 339–351.
- Emam, W.M., Saad, A.A., Riad, R. and AlWerfaly, H.A. (2014). Morphometric study and length-weight relationship on the squid *Loligo forbesi* (Cephalopoda: Loliginidae) from the Egyptian Mediterranean waters. *International Journal of Environmental Science and Engineering*, 5: 1–13.
- Ferguson, A., Mason, F.M. (1981) Allozyme evidence for reproductively isolated sympatric populations of brown trout *Salmo trutta* L. in Lough Melvin, Ireland. *J. Fish Biol.*, 18: 629–642.
- Fjøsne, K. and Gjøsæter, J. 1996. Dietary composition and the potential of food competition between 0-group cod (*Gadus morhua* L.) and some other fish species in the littoral zone. – ICES Journal of Marine Science, 53: 757–770.
- Frid, C.L.J., Clark, R.A. & Percival, P. (2001). How far have the ecological effects of fishing in the North Sea ramified? - *Senckenbergiana marit.*, 31 (2): 313–320.

- Frid, C.L.J. and Hall, S.J. (1999) Inferring changes in North Sea benthos from fish stomach analysis. *Marine Ecology-Progress Series*, 184: 183-188.
- Garner, P. (1996) Microhabitat use and diet of 0+ cyprinid fishes in a lentic, regulated reach of the River Great Ouse, England. *Journal of Fish Biology*, 48: 367-382.
- Garrod, C. and Harding, D. (1981) Predation by fish on the pelagic eggs and larvae of fishes spawning in the west central North Sea in 1976. *ICES CM* 1981/L:11.
- Garner, P. (1996) Microhabitat use and diet of 0+ cyprinid fishes in a lentic, regulated reach of the River Great Ouse, England. *Journal of Fish Biology*, 48: 367-382.
- Garner, P. (1996) Diel patterns in the feeding and habitat use of 0-group fishes in a regulated river: the River Great Ouse, England. *Ecology of Freshwater Fish*, 5: 175-182.
- Gibbons M.M. and McCarthy, T.K. (1984) Growth, maturation and survival of frogs *Rana temporaria* L. *Holarctic Ecology*, 7: 419-427.
- Graham, T.R. & Jones, J.W. (1962) The biology of Llyn Tegid trout 1960. *Proceedings of the Zoological Society of London*, 139(4): 657-683.
- Haram & Jones (1971) Some observations on the food of the gwyniad *Coregonus clupeoides pennantii* Valenciennes of Llyn Tegid (Lake Bala), North Wales. *Journal of Fish Biology*, 3: 287-295.
- Hellawell, J.M. (1971) The autecology of the chub, *Squalius cephalus* (L.), of the River Lugg and the Afon Llynfi. *Freshwater Biology*, 1(4): 369-387.
- Hellawell, J.M. (1971) The food of the grayling *Thymallus thymallus* (L.) of the River Lugg, Herefordshire. *Journal of Fish Biology*, 3(2): 187-197.
- Hellawell, J.M. (1972) The growth, reproduction and food of the roach *Rutilus rutilus* (L.), of the River Lugg, Herefordshire. *Journal of Fish Biology*, 4(4): 469-486.
- Hellawell, J.M. (1974) The ecology of populations of dace, *Leudscus leuciscus* (L.), from two tributaries of the River Wye, Herefordshire, England. *Freshwater Biology*, 4(6): 577-604.
- Hislop, J.R.G. & Youngson, A.F. (1984) A note on the stomach contents of salmon caught by longline north of the Faroe Islands in March, 1983. *ICES CM* 1984/14:17.
- Holden, M.J. (1966) The Food of the Spurdog, *Squalus acanthias* (L.). *J. Cons. perm. int. Explor. Mer.* 30(2): 255-266.
- Holden, M.J. and Tucker, R.N. (1974) The food of *Raja clavata* Linnaeus 1758, *Raja montagui* Fowler 1910, *Raja naevus* Muller and Henle 1841 and *Raja brachyura* Lafont 1873 in British waters. *J. Cons. perm. int. Explor. Mer.* 35(2): 189-193.
- Hop, H., Gjørseter, J., Danielssen, D.S. (1992) Seasonal feeding ecology of cod (*Gadus morhua* L.) on the Norwegian Skagerrak coast. *ICES Journal of Marine Science*, 49: 453-461. [values measured off frequency of occurrence graphs]
- Hunt, G. (2021). Spatial and Long-term Variability in Demersal Fish Species of the Western North Sea. PhD Thesis, School of Natural and Environmental Sciences Newcastle University. 208pp.
<https://theses.ncl.ac.uk/jspui/bitstream/10443/5577/1/Hunt%2C%20Georgina%20Louise%20Final%20Submission%20e-copy.pdf>
- Hunt, G., Cobain, M.R.D., McGill, R.A.R., Pinnegar, J.K., Engelhard, G.H., Wigham, B.D., Polunin, N.V.C. (2023) Quantifying spatio-temporal consistency in the trophic ecology of two sympatric flatfishes. *Journal of Fish Biology*, 103:378-392.
- Hutchinson, P. (1983) The ecology of smelt, *Osmerus eperlanus* (Linnaeus), from the River Thames and the River Cree. PhD Thesis, University of Edinburgh, 264pp.
- Jacobsen, J.A. & Hansen, L.P. (1996) The food of Atlantic salmon, *Salmo salar* L., north of the Faroe Islands. *ICES CM* 1996/M:10.
- Jones, J.R.E (1951) An Ecological Study of the River Towy. *Journal of Animal Ecology*, 20: 68-86.
- Kastelein, R.A. and van Battum, R. (1990) The relationship between body weight and morphological measurements in harbour porpoises (*Phocoena phocoena*) from the North Sea. *Aquatic Mammals*, 16(2): 48-52.

- Klimpel, S., Seehagen, A., Palm, H.W. (2003) Metazoan parasites and feeding behaviour of four small-sized fish species from the central North Sea. *Parasitol. Res.*, 91: 290–297.
- Last, J.M. (1979) The food of larval turbot *Scophthalmus maximus* L. from the west central North Sea. *J. Cons. int. Explor. Mer*, 38(3): 308–313.
- Lagardere, J.P. (1987) Feeding ecology and daily food consumption of common sole, *Solea vulgaris* Quensel, juveniles on the French Atlantic coast. *Journal of Fish Biology*, 30: 91–104.
- Liao, Yih-Yia (1999) Fisheries ecology of common wolffish, *Anarhichas lupus*, in the North Sea, Durham theses, Durham University. Available at Durham E-Theses Online: <http://etheses.dur.ac.uk/4403/>
- Lyle, J.M. (1983) Food and feeding habits of the lesser spotted dogfish, *Scyliorhinus canicula* (L.), in Isle of Man waters. *Journal of Fish Biology*, 23: 725–737.
- Nedreaas, K. (1985) Food and feeding habits of young saithe, *Pollachius virens* (L.), on the coast of western Norway. *ICES C.M.* 1985/G:22.
- Maes, J. & Ollevier, F. (2002) Size structure and feeding dynamics in estuarine clupeoid fish schools: field evidence for the school trap hypothesis. *Aquatic Living Resources*, 15: 211–216.
- Macer, C.T. (1977) Some aspects of the biology of the horse mackerel [*Trachurus trachurus* (L.)] in waters around Britain. *Journal of Fish Biology*, 10: 51–62.
- Magnussen, E. (2011) Food and feeding habits of cod (*Gadus morhua*) on the Faroe Bank. *ICES Journal of Marine Science*, 68(9): 1909–1917.
- Maitland, P.S. (1965) The feeding relationships of Salmon, Trout, Minnows, Stone Loach and Three-Spined Stickle-Backs in the River Endrick, Scotland. *Journal of Animal Ecology*, 34(1): 109–133.
- Mann, R.H.K, Bass, J.A.B, Leach, D., Pinder, A.C. (1997) Temporal and spatial variations in the diet of 0 group roach (*Rutilus rutilus*) larvae and juveniles in the River Great Ouse in relation to prey availability. *Regulated Rivers: Research & Management*, 13: 287–294.
- Merella, P. and Quetglas, A., Alemany, F. and Carbonell, A. (1997) Length-weight relationship of fishes and cephalopods from the Balearic Islands (Western Mediterranean). *Naga, the ICLARM Quarterly*, 20(3-4): 66–68.
- Marianne, R., Pawlowski, L., Kopp, D. (2022) TrophicCS: Spatialized trophic data of the Celtic Sea continental shelf food web. *Ecology*, 103: e3708. doi.org/10.1002/ecy.3708
- Markussen, N.H., Bjørge, A., Øritsland, N.A. (1989) Growth in harbour seals (*Phoca vitulina*) on the Norwegian coast. *Journal of Zoology*, 219: 433–440.
- Mattson, S. (1981) The food of *Galeus melastomus*, *Gadidulus argenteus thori*, *Trisopterus esmarkii*, *Rhinonemus cimbricus*, and *Glyptocephalus cynoglossus* (Pisces) caught during the day with shrimp trawl in a West-Norwegian fjord. *Sarsia* 66: 109–127.
- Mattson, S. (1990) Food and feeding habits of fish species over a soft sublittoral bottom in the Northeast Atlantic. 1. Cod (*Gadus morhua* L.) (Gadidae). *Sarsia* 75:247–260.
- Mattson, S. (1992) Food and feeding habits of fish species over a soft sublittoral bottom in the Northeast Atlantic. 3. Haddock (*Melanogrammus aeglefinus* (L.)) (Gadidae). *Sarsia* 77:33–45.
- McGoran, A.R., Clark, P.F., Morritt, D. (2016) Presence of microplastic in the digestive tracts of European flounder, *Platichthys flesus*, and European smelt, *Osmerus eperlanus*, from the River Thames. *Environmental Pollution*, 220(A): 744–751.
- Mendonca, V.M., Raffaelli, D., Boyle, P., Emes, C. (2007) The ecological role of overwintering fish in the food web of the Culbin Sands lagoon ecosystem, NE Scotland: Identifying major trophic links and testing effects of the fish *Pomatoschistus microps* (Pallas) on benthic invertebrates. *Scientia Marina*, 71 (4): 649–660.
- Menon, M.D. (1950) Bionomics of the poor-cod (*Gadus minutus* L.) in the Plymouth Area. *Journal of the Marine Biological Association of the United Kingdom*, 29(1): 185–239.

- Moriarty, R., Buitenhuis, E.T., Le Quéré, C. and Gosselin, M.-P. (2013) Distribution of known macrozooplankton abundance and biomass in the global ocean. *Earth System Science Data*, 5:241–257
- Ogilvie, H. (1927) Observations on the food of post-larval herring from the Scottish coast. Scientific Investigations of the Fishery Board for Scotland, 1927(1), 11pp.
- Ogilvie, H. (1938). The food of post-larval haddock with reference to the annual fluctuations in the haddock broods. Rapports et procès-verbaux des réunions (1903–1991). Volume 107. <https://doi.org/10.17895/ices.pub.9289>
- Ortiz de Zárate, V. and Cort, J.L. (1986) Stomach contents of immature bluefin tuna in the Bay of Biscay (Region 3 of EEC). ICES C.M. 1986/H:26
- Patel, R., Roy, S., Capuzzo, E., van der Kooij, J. (2023) Seasonality of diet overlap among small pelagic fish in the waters southwest of the UK. *Marine Ecology-Progress Series*, 708: 101-123.
- Pentelow, F.T.K. (1932). The food of the brown trout (*Salmo trutta* L.). *Journal of Animal Ecology*, 1(2): 101-107.
- Pentelow, F.T.K., Southgate, B.A., Bassindale, R. (1933) II The proportion of the sexes and the food of smolts of salmon and sea trout in the Tees Estuary. *Fishery Investigations, Series I., Vol III, Number 4*, 11-14.
- Pinnegar, J.K., Trenkel, V.M., Tidd, A.N., Dawson, W.A. and Du buit, M.H. (2003) Does diet in Celtic Sea fishes reflect prey availability? *Journal of Fish Biology*, 63 (S1): 197-212.
- Pisanu, B., Chapuis, J.L., Périn, R. (2003) *Syphacia obvelata* infections and reproduction of male domestic mice *Mus musculus domesticus* on a sub-Antarctic Island. *Journal of Helminthology* 77(3):247-53.
- Pitois, S.G. and Fox, C.J. (2006) Long-term changes in zooplankton biomass concentration and mean size over the Northwest European shelf inferred from Continuous Plankton Recorder data. *ICES Journal of Marine Science*, 63: 785-798.
- Rae, B.B. (1956) The food and feeding habits of the lemon sole. *Dep. Agric. Fish. Scotland, Marine Res.* 1956, No.3 pp.32.
- Rae, B.B. (1967) The food of cod in the North Sea and on West of Scotland grounds. *Dep. Agric. Fish. Scotland, Marine Res.* 1967, No.1 pp.68.
- Rae, B.B. (1967b) The food of cod on Faroese grounds. *Dep. Agric. Fish. Scotland, Marine Res.* 1967, No.6 pp.23.
- Rae, B.B. & Shelton, R.G.J. (1982) Notes on the food of nine species of elasmobranch (Part I) and nine species of demersal teleost (Part II) fishes from Scottish waters. *ICES CM* 1982/G:56.
- Robb, A.P. (1981) Observations on the food and diel feeding behaviour of pelagic 0-group gadoids in the northern North Sea *Journal of Fish Biology*, 18: 183-194.
- Robb, A.P., Hislop, J.R.G. (1980) The food of five gadoid species during the pelagic 0-group phase in the northern North Sea. *Journal of Fish Biology*, 16: 199-217.
- Rogers, S.I (1991) The occurrence of young scaldfish *Arnoglossus latevna* (Walbaum, 1792) in the eastern Irish Sea. *Journal of Fish Biology*, 39: 889-891.
- Rijnsdorp, A.D., Vingerhoed, B. (2001) Feeding of plaice *Pleuronectes platessa* L. and sole *Solea solea* (L.) in relation to the effects of bottom trawling. *Journal of Sea Research*, 45: 219-229.
- Sarno, B., Glass, C.W., Smith, G.W. (1994) Differences in diet and behaviour of sympatric saithe and pollack in a Scottish sea loch. *Journal of Fish Biology*, 45 (Supplement A): 1-11.
- Sayer, M.D.J., Gibson, R.N., Atkinson, R.J.A (1995) Growth, diet and condition of goldsinny on the west coast of Scotland. *Journal of Fish Biology*, 46: 317-340.
- Scharf, F.S., Juanes, F. and Rountree, R.A. (2000) Predator size - prey size relationships of marine fish predators: interspecific variation and effects of ontogeny and body size on trophic-niche breadth. *Marine Ecology-Progress Series*, 208: 229–248.

- Seisay, M.B.D (2001) Population Ecology of Dab (*Limanda limanda* L.) in the Eastern Irish Sea, North Wales. PhD Thesis, University of Wales, Bangor. 290pp.
- Sinha, V.R.P and Jones, J.W. (1967) On the food of the freshwater eels and their feeding relationship with the salmonids. *J. Zool., Lond.* 153: 119-137.
- Silva J.F., Ellis J.R. and Ayers R.A. (2013). Length-weight relationships of marine fish collected from around the British Isles. Science Series Technical Report, Cefas Lowestoft, 150: 109 pp.
- Spence, M.A., Dolder, P.J., Nash, R. and Thorpe, R.B. (2021) The use of a length-structured multispecies model fitted directly to data in near-real time as a viable tool for advice. *Frontiers in Marine Science*, 8:700534.doi: 10.3389/fmars.2021.700534.
- Studier, E.H. and Sevick, S.H. (1992) Live mass, water content, nitrogen and mineral levels in some insects from south-central lower Michigan. *Comparative Biochemistry and Physiology Part A: Physiology*, 103(3): 579-595.
- Sunnana, K. (1984) Stomach contents of cod, haddock and saithe on the M0re coast in 1982 and 1983. *ICES C.M.* 1984/G:56.
- Thijssen, R., Lever, A.J., Lever, J. (1974) Food composition and feeding periodicity of 0-group plaice (*Pleuronectes platessa*) in the tidal area of a sandy beach. *Netherlands Journal of Sea Research*, 8(4): 369-377.
- Thompson, M.S.A., Pontalier, H., Spence, M.A., Pinnegar, J.K., Greenstreet, S.P.R., Moriarty, M., Hélaouët, P., Lynam, C.P.(2020) A feeding guild indicator to assess environmental change impacts on marine ecosystem structure and functioning. *Journal of Applied Ecology*, 57(9): 1769-1781.
- Thompson, W. (1838b) XXXVIII - On fishes new to Ireland. *Annals of Natural History*. 1: 348-359.
- Townhill, B.L., Holt, R.E., Bogstad, B., Durant, J.M., Pinnegar, J.K., Dolgov, A.V., Yaragina, N.A., Johannesen, E., Ottersen, G. (2021) Diets of the Barents Sea cod (*Gadus morhua*) from the 1930s to 2018. *Earth Syst. Sci. Data*, 13, 1361–1370. doi.org/10.5194/essd-13-1361-2021.
- Tully, O. & O Ceidigh, P.(1989) The ichthyoneuston of Galway Bay (west of Ireland). II. Food of post-larval and juvenile neustonic and pseudo-neustonic fish. *Marine Ecology-Progress Series*, 51: 301-310.
- Victoria, O., Fayeye, T. R., Ayorinde, K., and Olojede, H. (2013). Relationship between body weight and linear body measurements in Japanese Quail (*Coturnix coturnix japonica*). *Journal of Scientific Research*, 6(1): 175-183.
- Weatherly, N.S. (1987) The diet and growth of 0-group dace, *Leuciscus leuciscus* (L.), and roach, *Rutilus rutilus* (L.), in a lowland river. *Journal of Fish Biology*, 30: 237-247.
- Wetsteijn, B. (1981) Feeding of North Sea turbot and brill. *ICES C.M.* 1981/G:74.

6 Appendices

6.1 Appendix 1: Database Field Descriptions

'Hauls' Table

1. `cruise-name` - usually includes the name of the vessel (or sampling programme) together with a year code (forms the linking variable to the 'PROVENANCE' table)
2. `Station` - the station number or area indicated in the original data. Where multiple station numbers are cited, or ICES rectangles then an over-arching letter and number is used and more details are provided in the 'comments' column.

3. haul_id - a unique identifier that is also replicated in the 'PREDATOR' database table (and forms the linking variable). It comprises the cruise name and station number.
4. date - sampling date with the format dd/mm/yyyy.
5. year - year in which samples were collected.
6. month - month in which samples were collected.
7. day - day of the month in which samples were collected.
8. quarter - quarter of the year in which samples were collected.
9. ices_rectangle - a sea area covering 0.5° Latitude, 1° Longitude used by the International Council for the Exploration of the Sea. For areas on land, the names of counties is used.
10. ices_sub_area - a large spatial unit used by the International Council for the Exploration of the Sea. For inland areas the name 'FRESHWATER' is used.
11. ices_division - a spatial sub-unit used by the International Council for the Exploration of the Sea. The North Sea is divided into 3 divisions (IVa,b,c), whereas the Channel is divided into 2 (VIId,e), the Celtic Sea into 5 (VIIIf,g,h,j) and the Irish Sea is 1 (VIIa). For inland areas the name of the country is used.
12. ices_roundfish - a defunct spatial sub-unit used by the International Council for the Exploration of the Sea, used in some older stomach sampling campaigns.
13. Sea - a broad geographical area used for DAPSTOM reporting. Includes: North Sea, Irish Sea, W Ireland, Celtic Sea, Channel, Biscay etc.
14. shot_lat_dd - Latitude of the shooting position expressed as a decimal.
15. shot_Lon_dd - Longitude of the shooting position expressed as a decimal.
16. Haul_lat_dd - Latitude of the hauling position expressed as a decimal.
17. Haul_lon_dd - Longitude of the hauling position expressed as a decimal.
18. shot_tm - time of day (hh:mm) when the net was deployed.
19. haul_tm - time of day (hh:mm) when the net was hauled.
20. Shot_depth_m - bottom depth of shooting position (in metres).
21. comments - additional information about the haul or station, sometimes listing the group of stations or ICES rectangles included.

'PROVENENCE' table

1. cruise_name - This is used as a linking variable to the main 'HAULS' table.
2. uploaded - Indicates the iteration of the DAPSTOM database where the particular dataset was first introduced.
3. source_type - Derivation of the original source material, i.e. MAFF/CEFAS DATASET, MAFF/CEFAS REPORT, PARTNER, PAPER (non Cefas), REPORT (non Cefas), CHARTER, COMMERCIAL.
4. data_input - The person who digitised the particular dataset.
5. Data_derived_from - Detailed description of the original source material (e.g. logbooks, folders or full citation for the paper/report).

'ICES' Table

1. haul_id - a unique identifier that forms the linking variable to the 'HAULS' table. It comprises the cruise name and station number.
2. country - the country responsible for collecting the data. Format as specified in the 'ICES vocabulary' and DATRAS database (e.g. United Kingdom (England, France etc)).
3. country_code - 3 digit country code as specified in the 'ICES vocabulary' and DATRAS database (e.g. United Kingdom (ENG,SCO,FRA etc)).
4. ship_code - 3 or 4 digit ship code as specified in the 'ICES vocabulary' and DATRAS database, supplemented with additional codes for older vessels (forms the linking variable to the 'SHIPS' table)
5. gear_type - the sampling method used to catch the fish, usually some sort of trawl.
6. Yos_method - the gear type categorised into 3 digit groups used by ICES in the 'Year of the Stomach' database (e.g. DEM, DGN, DHL, PEL)
7. Ices_gear - 3 digit gear code as specified in the 'ICES vocabulary' and DATRAS database (e.g. BMT, GIL, GOV, PHT etc).

'SHIPS' Table

1. ship_code - 3 or 4 digit ship code as specified in the 'ICES vocabulary' and DATRAS database, supplemented with additional codes for older vessels (forms the linking variable to the 'ICES' table)
2. ship_name - Full name of the vessel used to collect the samples.

'PREDATOR' Table

1. haul_id - a unique identifier which is also replicated in the main 'HAULS' table (and forms the linking variable). It comprises the cruise name and station number.
2. pred_id - a unique identifier for the individual predator animal (or group of animals). This is needed because there is sometimes more than one prey item within a single stomach) (forms the linking variable to the 'PREY' table).
3. Pred - the predator species, indicated by a 3 digit code.
3. predator_tsn - Taxonomic Serial Number (TSN), a unique, persistent, non-intelligent identifier used internationally in the context of the Integrated Taxonomic Information System (ITIS) (forms the linking variable to the 'PREDATOR TAXONOMY' table).
4. pred_length_cm - the length (in cm) of the individual predator.
5. Pred_weight_g - weight of the individual predator (in grams).
6. pred_mat - maturity stage of the individual predator (as listed in the original paper source). Sometimes given a numerical value (spanning 1-7), or simply I = immature, M = mature, S = Spent, R = Running.
7. pooled - an indicator of whether or not the record represents a group of individuals or a single animal. y = yes (pooled), n = no (a single animal).
8. num_stomachs - indicates how many individuals the PRED ID represents. This has a value of '1' where the data has been collected at the individual animal level, however it may be higher for pooled data.
9. num_empty - indicates the number of empty stomachs included within the PRED ID.
10. size_category - used when the record represents a group of individuals, to indicate the size range included (if known).
11. mean_length_cm - used when the record represents a group of individuals, to indicate the average (or median) size (in cm).

12. `tpl` - Theoretical prey length based on the observed relationship between predator and prey lengths in Pinnegar et al. (2003) and Scharf et al. (2000).

‘PREDATOR TAXONOMY’ look-up table

1. `predator_tsn` - Taxonomic Serial Number (TSN), a unique, persistent, non-intelligent identifier used internationally in the context of the Integrated Taxonomic Information System (ITIS) (forms the linking variable to the ‘PREDATOR’ table).
2. `aphiaid` - Aphia code used internationally as a taxonomic identifier from the World Register of Marine Species (WoRMS, www.marinespecies.org).

‘PREY’ table

1. `id` - a unique identifier for each record (automatically added by the Postgres software).
2. `pred_id` - a unique identifier for the individual predator animal (or group of animals). This is needed because there is sometimes more than one prey item within a single stomach (forms the linking variable to the ‘PREDATOR’ table).
3. `prey_tsn` - Taxonomic Serial Number (TSN), a unique, persistent, non-intelligent identifier used internationally in the context of the Integrated Taxonomic Information System (ITIS). This is used as a linking variable to the ‘PREY TAXONOMY’ look-up table.
4. `prey_name` - prey type, as listed in the original paper source.
5. `Prey_length_cm` - length or size of individual prey item (in cm).
6. `ind_prey_wgt_g` - the weight of the individual prey item (in grams).
7. `Digestion` - standardised digestion stage as used by ICES in the ‘Year of the Stomach’ database (e.g. 0, 1, 2 and 9999 (unknown)).
8. `prey_qualifier_code` - codes used to differentiate different life stages (see table 1 of this report). This is used as a linking variable to the ‘PREY QUALIFIER’ look-up table.
9. `min_num` - where the prey number is given, this is reproduced in this column. Where no prey number is given, then a minimum of ‘1’ is indicated. Sometimes based on the frequency of occurrence, together with the number of (non-empty) stomachs.
10. `cpw` - Calculated Prey Weight. For invertebrate prey items the average prey weight (in the PREY TAXONOMY table) is multiplied by the number of prey consumed (in the PREY table). For fish prey items the ‘theoretical prey length’ from the PREDATOR table is used in conjunction with the weight-length exponents (a, b) from the PREY TAXONOMY table. This is then multiplied by the minimum number of prey consumed (in the PREY table).

‘PREY TAXONOMY’ look-up table

1. `prey_tsn` - Taxonomic Serial Number (TSN), a unique, persistent, non-intelligent identifier used internationally in the context of the Integrated Taxonomic Information System (ITIS). This is used as a linking variable to the ‘PREY’ table.
2. `aphiaid` - Aphia code used internationally as a taxonomic identifier from the World Register of Marine Species (WoRMS, www.marinespecies.org).
1. `is_fish` - indicating whether the prey item is a fish (y) or some other type (n).
2. `is_fish_a` - exponent in the weight-length relationship (for fish prey items only).

3. `is_fish_b` - exponent in the weight-length relationship (for fish prey items only).
4. `is_fish_source` - source of the weight-length exponents (e.g. Silva et al. (2013); Fishbase).
5. `av_invert_wgt_g` - Average weight of the prey item based on literature sources (for invertebrate and non-fish prey items).
6. `av_invert_source` - Literature source for the average weight estimate (see above).

'PREY QUALIFIER' look-up table

11. `prey_qualifier_code` - codes used to differentiate different life stages (see table 1 of this report). This is used as a linking variable to the 'PREY' table.
12. `qual_description` - Life stage or body part of prey item listed in the original source, e.g. Q1 = None, Q2 = larvae, Q4 = eggs etc. (see table 1 of this report).

6.2 Appendix 2: Number of records and stomachs by species

Species introduced for the first time in Version 6.3 of the DAPSTOM database, indicated in red.

PRED	SCIENTIFIC	COMMON	TSN CODE	RECORDS	STOMACHS
AAS	ALOSA ALOSA	ALLIS SHAD	166401	3	3
ACH	SALVELINUS ALPINUS	CHAR	166787	9	24
ALB	THUNNUS ALALUNGA	ALBACORE	161997	1	1
ALS	SQUATINA SQUATINA	ANGELSHARK (MONKFISH)	171854	22	19
ANE	ENGRAULIS ENCRASICOLUS	EUROPEAN ANCHOVY	171982	506	510
ANF	LOPHIIDAE	ANGLER FISHES	160882	28	26
ARG	ARGENTINIDAE	ARGENTINES	166705	41	23
ATH	ALOPIAS VULPINUS	THRESHER SHARK	159844	5	3
ATS	ALLOTEUTHIS SUBULATA	SQUID	164478	16	16
BBY	BLENNIUS OCELLARIS	BUTTERFLY BLENNY	166756	16	24
BFT	THUNNUS THYNNUS	BLUE-FIN TUNNY	164754	18	33
BIB	TRISOPTERUS LUSCUS	WHITING-POUT (BIB)	163581	973	1932
BKS	SPONDYLIOSOMA CANTHARUS	BLACK SEABREAM	166025	8	8
BLG	GOBIUS NIGER	BLACK GOBY	167050	88	458
BLL	SCOPHTHALMUS RHOMBUS	BRILL	172881	360	1333
BLR	RAJA BRACHYURA	BLONDE RAY	172043	144	215
BNW	LABRUS BERGYLTA	BALLAN WRASSE	159772	185	349
BOF	CAPROS APER	BOAR FISH	172806	304	288
BON	SARDA SARDA	BONITO	164482	1	1
BRT	MYOXOCEPHALUS SCORPIUS	BULLROUT	162071	80	185
BSH	PRIONACE GLAUCA	BLUE SHARK	161722	76	353
BSK	CETORHINUS MAXIMUS	BASKING SHARK	173026	11	2
BTF	PHOLIS GUNNELLUS	BUTTER FISH	163613	414	725
BUL	COTTUS GOBIO	BULLHEAD	171978	6	15
CAA	ANARHICHAS LUPUS	CATFISH (WOLFFISH)	163761	239	674
CAC	LEPADOGASTER CANDOLII	CONNEMARRA CLINGFISH	164706	35	35
CAJ	ANARHICHAS DENTICULATUS	JELLY CATFISH	171676	1	1
CAP	MALLOTUS VILLOSUS	CAPELIN	171974	31	1
CAS	ANARHICHAS MINOR	SPOTTED CATFISH	636467	9	2
CDT	CALLIONYMUS LYRA	COMMON DRAGONET	170739	723	1748
CLG	CRYSTALLOGOBIUS LINEARIS	CRYSTAL GOBY	167578	4	4
COD	GADUS MORHUA	COD	161708	75923	79035
COE	CONGER CONGER	EUROPEAN CONGER EEL	162575	91	79
CTM	COTTUNCULUS MICROPS	POLAR SCULPIN	623503	2	1
CUB	LEUCISCUS CEPHALUS	CHUB	162022	172	895
CUR	LEUCORAJA NAEVUS	CUCKOO RAY	163663	583	979
CUW	LABRUS MIXTUS	CUCKOO WRASSE	170281	29	17
CWG	SYMPHODUS MELOPS	CORKWING	165324	269	1061
DAB	LIMANDA LIMANDA	DAB	161831	18597	24576
DBM	GALEUS MELASTOMUS	BLACKMOUTHED DOGFISH	161722	71	260

DCH	DALATIAS LICHA	DARKIE CHARLIE	551497	3	8
DCO	DELPHINUS DELPHIS	COMMON DOLPHIN	623496	5	2
DGN	SCYLORHINUS STELLARIS	NURSE HOUND	171302	184	145
DGS	SQUALUS ACANTHIAS	SPURDOG	171342	1277	2955
DTX	CALLIONYMIDAE	DRAGONETS	616613	26	60
EGR	MYLIOBATIS AQUILA	EAGLE RAY	166365	3	1
EKT	ZEUGLOPTERUS REGIUS	EKSTROMS TOPKNOT	160883	5	15
ELE	ANGUILLA ANGUILLA	EUROPEAN EEL	171126	1345	8302
ELP	ZOARCES VIVIPARUS	EELPOUT/VIVIPARUS BLENNY	162187	20	39
ESB	DICENTRARCHUS (MORONE) LABRAX	EUROPEAN SEABASS	162035	361	3106
FBK	ALBURNUS ALBURNUS	BLEAK	170335	48	112
FBM	ABRAMIS BRAMA	BREAM	168520	88	700
FDC	LEUCISCUS LEUCISCUS	DACE	161128	130	1067
FGN	GOBIO GOBIO	GUDGEON	166287	49	250
FGR	THYMALLUS THYMALLUS	GRAYLING	692071	201	424
FLE	PLATICHTHYS FLESUS	FLOUNDER (EUROPEAN)	170737	1250	6077
FLX	PLEURONECTIFORMES	FLATFISHES	643890	1	1
FMW	PHOXINUS PHOXINUS	MINNOW	168470	167	739
FPE	PERCA FLUVIATILIS	EUROPEAN PERCH	172409	1043	4873
FPI	ESOX LUCIUS	EUROPEAN PIKE	642613	861	3232
FRD	SCARDINIUS ERYTHROPTHALMUS	RUDD	170377	8	129
FRF	GYMNOCEPHALUS CERNUA	RUFFE	172888	49	238
FRO	RUTILUS RUTILUS	ROACH	171645	730	4005
FRR	ENCHELYOPUS CIMBRIUS	FOUR-BEARDED ROCKLING	163661	307	634
FSB	BLICCA (ALBURNUS) BJOERKNA	SILVER BREAM	170333	13	100
FTL	BARBATULA BARBATULA	STONE LOACH	168588	44	316
FVR	CILIATA MUSTELA	FIVE-BEARDED ROCKLING	167408	279	2002
GAD	GADIDAE	CODLIKE FISHES	164772	5	3
GAG	GALEORHINUS GALEUS	TOPE SHARK	170371	137	108
GAR	BELONE BELONE	GARFISH	614239	122	1360
GDY	CTENOLABRUS RUPESTRIS	GOLDSINNY	171746	141	506
GFB	PHYCIS BLENNOIDES	GREATER FORKBEARD	164780	60	59
GLH	REINHARDTIUS HIPPOGLOSSOIDES	GREENLAND HALIBUT	171699	15	13
GMG	POMATOSCHISTUS MICRIPS	COMMON GOBY	164756	34	55
GOB	GOBIDAE	GOBIES	163596	223	762
GPF	SYNGNATHUS ACUS	GREAT PIPEFISH	167454	16	30
GRF	SEBASTES NORVEGICUS	GOLDEN REDFISH	564149	3	3
GSE	HYPEROPLUS LANCEEOLATUS	GREAT SANDEEL	162139	357	738
GSK	SOMNIOSUS MICROCEPHALUS	GREENLAND SHARK	167390	12	4
GSS	ARGENTINA SILUS	GT SILVER SMELT	623510	2	2
GUG	EUTRIGLA GURNARDUS	GREY GURNARD	556692	8351	8894
GUL	ASPIRIGLA OBSCURA	LONG-FINNED GURNARD	555704	25	18
GUR	CHELIDONICHTHYS CUCULUS	RED GURNARD	160617	1081	1132
GUS	TRIGLOPORUS LASTOVIZA	STREAKED GURNARD	166356	28	19
GUX	TRIGLA SPP	GURNARDS	168590	607	320

HAD	MELANOGRAMMUS AEGLEFINUS	HADDOCK	164497	21785	21243
HAL	HIPPOGLOSSUS HIPPOGLOSSUS	HALIBUT	159819	23	12
HER	CLUPEA HARENGUS	HERRING	162001	7982	77469
HGF	MYXINE GLUTINOSA	HAGFISH	172835	23	144
HKE	MERLUCCIUS MERLUCCIUS	EUROPEAN HAKE	162057	2626	12116
HMM	TRACHURUS MEDITERRANEUS	MEDITERRANEAN SCAD	564134	20	12
HOM	TRACHURUS TRACHURUS	HORSE-MACKEREL (SCAD)	171691	1383	1103
ISF	ARNOGLOSSUS IMPERIALIS	IMPERIAL SCALDFISH	161813	44	43
ISSH	CLUPEA HARENGUS	HERRING (ICELANDIC SPRING SPAWNING)	167046	702	399
JOD	ZEUS FABER	JOHN DORY	166781	483	499
LBI	LEPIDORHOMBUS BOSCHII	FOUR SPOT MEGRIM	172834	266	250
LCT	BRANCHIOSTOMA LANCEOLATUM	LANCET	616195	1	1
LEM	MICROSTOMUS KITT	LEMON SOLE	172749	2316	6723
LFB	RANICEPS RANINUS	LESSER FORKBEARD	160900	1	1
LIN	MOLVA MOLVA	COMMON LING	166464	172	506
LLV	LOLIGO VULGARIS	SQUID	161789	6	6
LPE	LEPIDION EQUES	MORID COD	160651	41	1281
LSD	SCYLIORHINUS CANICULA	LESSER SPOTTED DOGFISH	167318	2882	4734
LSG	THOROGOBIOUS EPHIPIATUS	LEOPARD SPOTTED GOBY	171677	42	115
LSS	ARGENTINA SPHYRAENA	LSR SILVER SMELT	160181	433	559
LUM	CYCLOPTERUS LUMPUS	LUMPSUCKER	159690	219	250
LZG	POMATOSCHISTUS LOZANOI	LOZANO'S GOBY	170733	37	537
MAC	SCOMBER SCOMBRUS	(EUROPEAN) MACKEREL	164755	10636	7540
MBY	CORYPHOBLENNIUS GALERITA	MONTAGUE'S BLENNY	160713	311	3
MEG	LEPIDORHOMBUS WHIFFIAGONIS	MEGRIM	167255	4311	3924
MGN	LIZA AURATA	GOLDEN MULLET	164740	1	1
MON	LOPHIUS PISCATORIUS	ANGLERFISH (MONK)	550561	1359	3168
MSE	AMMODYTES MARINUS	SANDEEL	160067	118	1193
MSS	LIPARIS MONTAGUI	MONTAGUE'S SEASNAIL	166595	9	8
MTG	MUGIL CEPHALUS	GREY MULLET	161716	4	3
MTL	CHELON (CRENIMUGIL) LABROSUS	THICK LIPPED MULLET	166326	62	52
MUL	MUGILIDAE	GREY MULLET	164765	6	3
MUR	MULLUS SURMULETUS	RED MULLET	172414	191	155
MYX	MYCTOPHIDAE	LANTERN FISHES	170290	7	36
NKT	ZEUGLOPTERUS NORVEGIUS	NORWEGIAN TOPKNOT	160611	82	485
NNR	CILIATA SEPTENTRIONALIS	NORTHERN ROCKLING	162064	58	470
NOP	TRISOPTERUS ESMARKI	NORWAY POUT	159916	1103	9099
NPF	SYNGNATHUS ROSTELLATUS	NILSSON'S PIPEFISH	164727	18	28
NSSH	CLUPEA HARENGUS	HERRING (NORWEGIAN SPRING SPAWNING)	163580	2199	962
OPA	LAMPRIS GUTTATUS	OPAH (MOON-FISH)	160901	3	2
PFX	SYNGNTHIDAE	PIPE-FISHES/SEAHORSES	165284	3	1
PIL	SARDINA PILCHARDUS	PILCHARD	623023	1065	609
PLA	HIPPOGLOSSOIDES PLATESSOIDES	AMERICAN PLAICE (LR DAB)	161341	846	2529
PLE	PLEURONECTES PLATESSA	EUROPEAN PLAICE	172419	19282	44406
PLN	COREGONUS LAVARETUS	POLLAN	164795	142	333

PLS	MAUROLICUS MUELLERI	PEARLSIDE	173002	12	8
POA	BRAMA BRAMA	RAYS BREAM (POMFRET)	173051	12	8
POC	BOREOGADUS SAIDA	ARCTIC COD	164502	1	1
POD	TRISOPTERUS MINUTUS	POOR COD	171850	1676	6783
POG	AGONUS CATAPHRACTUS	POGGE (ARMED BULLHEAD)	164760	167	231
POK	POLLACHIUS VIRENS	SAITHE	160880	3203	6494
POL	POLLACHIUS POLLACHIUS	POLLACK	160034	560	1298
POR	LAMNA NASUS	PORBEAGLE SHARK	173021	18	7
PRP	PHOCOENA PHOCOENA	HARBOUR PORPOISE	170992	3	1
PTG	POMATOSCHISTUS PICTUS	PAINTED GOBY	163983	30	616
PTR	RAJA MICROOCELLATA	PAINTED RAY	771137	115	1279
RBF	CHIMAERA MONSTROSA	RABBIT FISH(RAT-TAIL)	164728	25	28
RBM	HELICOLENUS DACTYLOPTERUS	BLUE-MOUTH REDFISH	166463	58	53
REB	SEBASTES MENTELLA	ROSE FISH	160242	9	9
RED	SEBASTES SPP	REDFISHES	172902	10	42
REG	SEBASTES MARINUS	NORWAY HADDOCK	166443	46	51
RKG	GOBIUS PAGANELLUS	ROCK GOBY	171682	326	487
RNF	REGALECUS GLENE	RIBBON FISH	630409	1	1
ROL	PHYCINAE	ROCKLINGS	161022	18	26
RPF	CEPOLA MACROPHTHALMA	RED BANDFISH	160959	49	13
SAL	SALMO SALAR	N.ATLANTIC SALMON	172421	785	7866
SAN	AMMODYTES SPP	SANDEELS	164748	1409	672
SAR	RAJA CIRCULARIS	SANDY RAY	564126	11	10
SBY	LUMPENUS LAMPRETAEFORMIS	SNAKE BLENNY	159911	62	28
SCF	LEPADOGASTER LEPADOGASTER	SHORE CLINGFISH	161722	31	53
SDF	ARNOGLOSSUS LATERNA	SCALD FISH	170317	256	1034
SDG	POMATOSCHISTUS MINUTUS	SAND GOBY	160983	153	1102
SDR	RAJA MONTAGUI	SPOTTED RAY	564143	692	1830
SDS	MUSTELUS ASTERIAS	STARRY SMOOTH HOUND	169418	116	94
SDT	CALLIONYMUS MACULATUS	SPOTTED DRAGONET	180473	23	28
SGR	DASYATIS PASTINACA	STING RAY	164774	2	1
SGS	HEXANCHUS GRISEUS	SIX-GILLED SHARK	636404	12	12
SHR	RAJA FULLONICA	SHAGREEN RAY	167612	88	166
SHY	LIPOPHRYS PHOLIS	SHANNY	164712	1360	2671
SKA	RAJIDAE	SKATES AND RAYS	164777	125	236
SKT	DIPTURUS BATIS	COMMON SKATE	166972	212	393
SLY	COREGONUS LAVARETUS	SHELLY	161950	9	10
SME	OSMERUS EPERLANUS	SMELT(SPARKLING)	616605	39	1085
SMH	MUSTELUS MUSTELUS	SMOOTH HOUND	161996	37	17
SMT	ATHERINA PRESBYTER	SAND SMELT	172805	9	16
SNI	MACRORHAMPHOSUS SCOLOPAX	SNIPE-FISH	172829	4	2
SNP	NEROPHIS OPHIDION	STRAIGHT-NOSED PIPEFISH	164758	1	1
SOL	SOLEA SOLEA (S.VULGARIS)	SOLE (DOVER SOLE)	159907	1916	4069
SOS	PEGUSA LASCARIS	SAND SOLE	171671	29	17
SOT	BUGLOSSIDIUM LUTEUM	SOLENETTE	165594	81	541
SPR	SPRATTUS SPRATTUS	SPRAT	166320	4801	5095

SRR	GAIDROPSARUS MEDITERRANEUS	SHORE ROCKLING	164501	32	121
SRT	MALACOCEPHALUS LAEVIS	SOFT HEADED RATTAIL	164766	6	5
SSL	LIPARIS LIPARIS	SEA SNAIL	167578	45	115
SSN	TAURULUS BUBALIS	SEA SCORPION	163580	538	942
SSS	SPINACHIA SPINACHIA	SEA STICKLEBACK	160845	15	12
SVS	HEPTRANCHIAS PERLO	SHARPNOSE SEVEN-GILL SHARK	172873	1	1
SYP	GADICULUS ARGENTEUS	SILVERY POUT	163658	114	902
SYR	AMBLYRAJA RADIATA	STARRY RAY	172930	370	522
SYS	ECHINORHINUS BRUCUS	SPINY SHARK	166593	2	1
TAS	ALOSA FALLAX	TWAITE SHAD	164701	109	597
TBR	GAIDROPSARUS VULGARIS	THREE-BEARDED ROCKLING	172933	90	356
TBS	MICROCHIRUS VARIEGATUS	THICKBACK SOLE	164751	163	536
TBY	PARABLENNIUS GATTORUGINE	TOMPOT BLENNY	166745	13	20
TFX	SALMO FEROX	FEROX TROUT	160889	8	17
TGL	SALMO STOMACHICUS	GILLAROO TROUT	171980	10	27
THR	RAJA CLAVATA	THORNBACK RAY (ROKER)	160424	2421	4364
TKT	ZEUGOPTERUS PUNCTATUS	TOPKNOT	623425	11	40
TPG	APHIA MINUTA	TRANSPARENT GOBY	165389	3	5
TRS	SALMO TRUTTA	SEA TROUT (BROWN TROUT)	169229	2718	7304
TSC	DIPLECOGASTER BIMACULATA	TWP SPOTTED CLINGFISH	162039	10	10
TSE	AMMODYTES TOBIANUS	SANDEEL	164744	133	364
TSG	GOBIUSCULUS FLAVESCENS	TWO-SPOT GOBY	172894	144	822
TSN	SALMO NIGRIPINNIS	SONAGHEN TROUT	160240	9	15
TSS	GASTEROSTEUS ACULEATUS	THREE-SPINED STICKLEBACK	637881	37	160
TUB	CHELIDONICHTHYS LUCERNUS	TUB GURNARD	167044	374	463
TUR	PSETTA MAXIMA	TURBOT	171341	862	3621
UNR	RAJA UNDULATA	UNDULATE RAY	172033	7	2
USK	BROSME BROSME	TUSK	160065	31	220
VLP	LYCODES VAHLII	VAHL'S EELPOUT	631023	12	17
WAF	LOPHIUS BUDEGASSA	BLACK-BELLIED ANGLERFISH	82375	276	225
WEG	TRACHINUS DRACO	GREATER WEEVER FISH	172877	7	43
WEL	ECHIICHTHYS VIPERA	LESSER WEEVER FISH	171971	2650	2709
WHB	MICROMESISTIUS POUTASSOU	BLUE WHITING	171698	1799	2733
WHG	MERLANGIUS MERLANGUS	WHITING	160788	47929	36184
WIT	GLYPTOCEPHALUS CYNOGLOSSUS	WITCH	172702	433	808
WPF	NEROPHIS LUMBRICIFORMIS	WORM PIPEFISH	164690	24	223

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Centre for Environment Fisheries & Aquaculture Science



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Our environmental monitoring and assessment programmes are fundamental to the sustainable development of marine and freshwater industries.

Through the application of our science and technology, we play a major role in growing the marine and freshwater economy, creating jobs, and safeguarding public health and the health of our seas and aquatic resources

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- industries across a range of sectors including offshore renewable energy, oil and gas emergency response, marine surveying, fishing and aquaculture.
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